FAIR workflows to chart and characterize the conformational landscape of native proteins.

A combined work of **ELIXIR 3D-BioInfo** structural community and the **BioExcel** Centre of Excellence for Computational Biomolecular Research

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Proteins are **dynamic** systems that adopt multiple **conformational states**, a property essential for many **biological processes** (e.g. binding other proteins, nucleic acids, small molecule ligands, or switching between functionally active and inactive states). Characterizing the different **conformational states** of proteins and the transitions between them is therefore critical for gaining insight into their **biological function** and can help explain the effects of genetic variants in **health** and **disease** and the action of drugs.

The **ELIXIR 3D-BioInfo** structural community gathers together experts in the field of **structural bioinformatics** providing methods and tools to analyse, predict, archive and validate the **three-dimensional (3D) structure** data of biomacromolecules such as **proteins**, **RNA** or **DNA**.

The **3D-BioInfo** Implementation Study ***Building on PDBe-KB to chart and characterize the conformational landscape of native proteins*** aims to create an infrastructure to chart the experimentally sampled **conformational diversity** of **native proteins** by exploiting data from the **PDB** and **state-of-the-art computational tools**. The project was designed from scratch to generate **FAIR workflows** with the integration of these tools.

The **BioExcel Centre of Excellence for Computational Biomolecular Research** has designed and developed a software library for interoperable and reproducible **biomolecular simulation workflows** (BioExcel Building Blocks, BioBB). The combination of the **BioBB** library and the **ELIXIR 3D-BioInfo Implementation Study** tools is producing a set of **FAIR workflows** for the generation of **conformational ensembles** and its **flexibility** analysis.

Findable

- Registered in **bio.tools** with associated metadata (inputs, outputs, reference, documentation, etc.)
- Registered in **WorkflowHub**, in different flavours (Jupyter, Python, CWL, Galaxy) and unique DOIs
- Tagged with **BioSchemas** as a Software Application, with information such as main URL, software type, source code repository in GitHub, citation, license and version.
- Monitored in **OpenEBench**: uptime, access time, versions, license, publications, documentations, etc.

Accessible

- Source code available in **GitHub**, with associated license (Apache 2.0)
- Documentation available in **ReadTheDocs**, with installation instructions
- Extensive Information available in the **BioBB homepage** workflow section

Interoperable

- The **BioBB** library is offering tool's **interoperability** by definition: is designed as a collection of **Python wrappers** on top of **biomolecular simulation tools** providing an **interoperable** architecture
- **Jupyter Notebooks** and **Galaxy** web-based analysis platforms offer the possibility to integrate the workflow with already existing tools and libraries in a very **easy, graphical** and **intuitive** way
- **Common Workflow Language (CWL)** descriptions are built upon an **open standard** for describing how to run command line **tools** and **connect** them to create **workflows**

Reusable

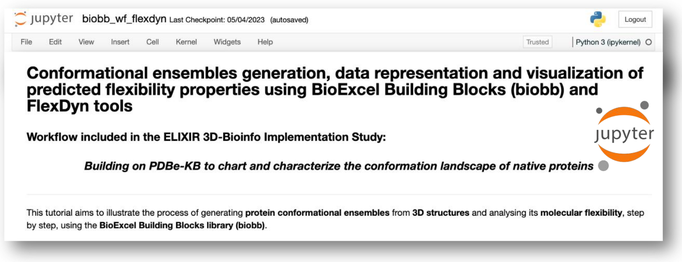
- Packaged in **BioConda** with a **workflow environment** including all dependencies with specific **program versions**, allowing the execution from **Python**, **Jupyter Notebooks** and **Galaxy**
- Tools used in the workflow steps are packaged in **BioContainers** (***Docker***, ***Singularity***), allowing the workflow execution from a **CWL** description or HPC-focused workflow managers (e.g. **PyCOMPSs**)
- **Workflows** and associated **BioBB tools** are licensed under the **Apache License version 2.0**, a permissive **free software** license.

Example of **BioBB-3D-BioInfo workflow**:

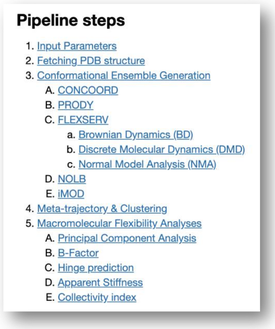
Conformational ensembles generation, data representation and visualization of predicted flexibility properties

Available as:

Jupyter Notebook
Galaxy workflow (https://biobb.usegalaxy.es/)
Common Workflow Language



Jupyter Notebook version of the Workflow

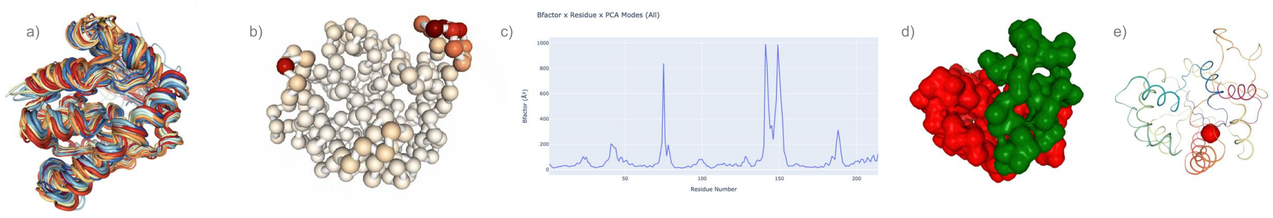


Steps included In the Workflow

Link to the workflow information

Conclusions and future work:

- **3D-BioInfo** has built its first **FAIR workflow** in collaboration with the **BioExcel CoE** and using the **BioBB** library
- **3D-BioInfo** will continue working on new **FAIR workflows**, presenting them to the **structural bioinformatics** field, and encouraging the community to follow **best practices** in **software/workflow development**



Workflow intermediate results using the Adenylate Kinase example (PDB Code: 1AKE): a) Generated Conformational Ensemble; b & c) Bfactor/Flexible domain representation from the generated ensemble; d) Flexible Domains prediction and e) Protein Hinges prediction

References:

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